CH04

- uniform sampling $t = nT_s$ $f_s = \frac{1}{T_s}$

$\therefore t = \frac{n}{f_s}$

- Shannon Nyquist Theorem (Sampling theorem)
reconstruct 조건은 $f_s > 2f_{max}$ 이어야 함

- Nyquist rate $f_s > 2f_{max}$

Discrete time sinusoid

$x(t) = A\cos(\omega t + \phi)$

↓

$x[n] = x(nT_s) = A\cos(\omega n T_s + \phi)$

$= A\cos(\hat{\omega} n + \phi)$

ωT_s ← x_n

↓

$\hat{\omega}$

$\therefore \hat{\omega} = \omega T_s = \frac{\omega}{f_s}$: digital frequency

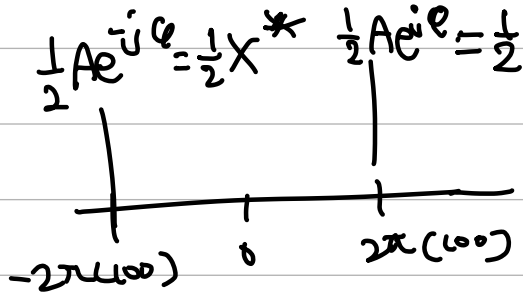
$\omega = 2\pi f$ $\therefore \hat{\omega} = \frac{2\pi f}{f_s}$

$\hat{\omega} = 0$ 이라면 $f = 0$

$\hat{\omega} = 2\pi$ 이라면 $f = f_s$

- 예제 $x(t) = A\cos(2\pi(100)t + \varphi)$

$$= \frac{1}{2}Ae^{j\varphi}e^{j2\pi(100)t} + \frac{1}{2}Ae^{-j\varphi}e^{-j2\pi(100)t}$$

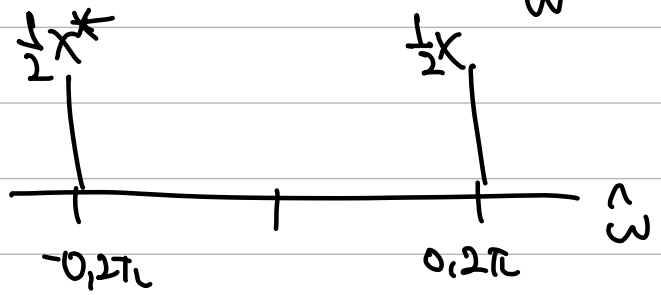


$(f_s = 1000Hz)$

$$x[n] = x\left(\frac{n}{f_s}\right) = A\cos\left(2\pi(100)\frac{n}{1000} + \varphi\right)$$

$$= A\cos(2\pi(0.1)n + \varphi)$$

$\hat{\omega}$



- Aliasing

- Normalized radian frequency

$$\hat{\omega} = \frac{2\pi f}{f_s} + 2\pi l$$

↘ folded alias

$$-\frac{2\pi f}{f_s} + 2\pi l$$

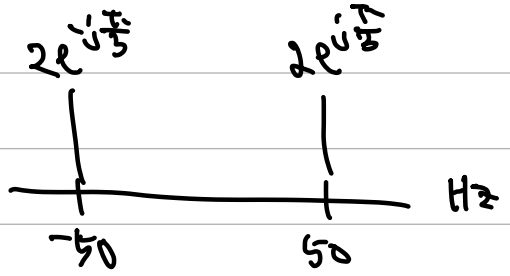
Normalized cyclic frequency

$$\hat{f} = \frac{f}{f_s}$$

- 3/2

$x(t) = 4\cos(2\pi(50)t + \frac{\pi}{3})$

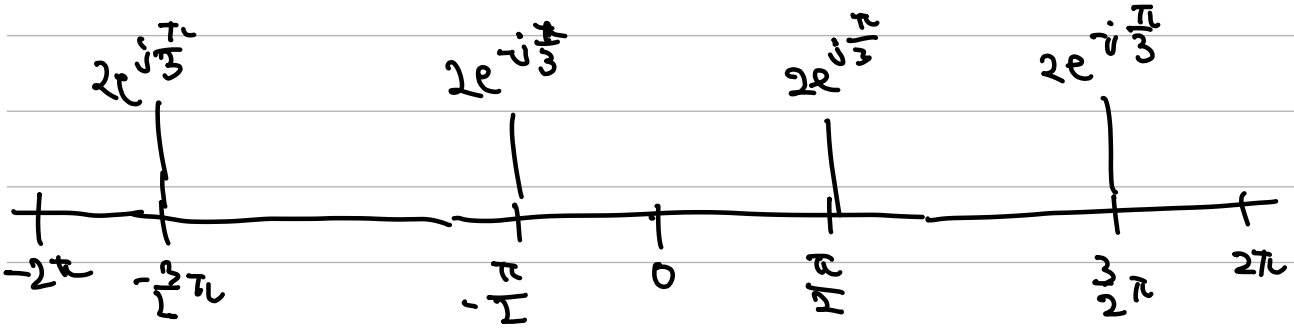
(a) $= 2e^{j\frac{\pi}{3}} e^{j2\pi(50)t} + 2e^{-j\frac{\pi}{3}} e^{-j2\pi(50)t}$



(b) $f_s = 200Hz$ $nT_s = \frac{n}{f_s} = t$

$x[n] = 4\cos(2\pi(50)\frac{n}{200} + \frac{\pi}{3})$

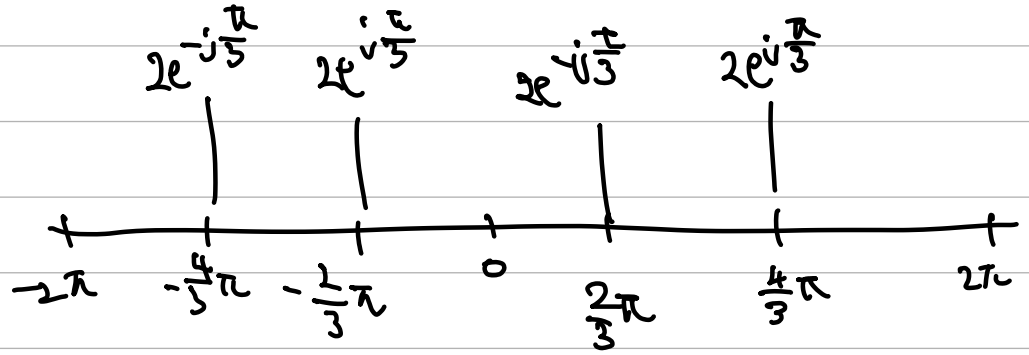
$\hat{\omega} = \frac{2\pi f}{f_s} = \frac{2\pi \cdot 50}{200} = \frac{\pi}{2}$



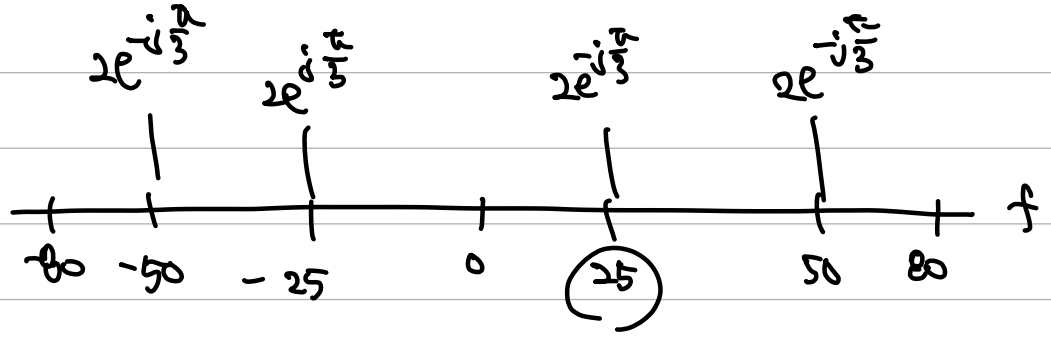
(c) $f_s = 75Hz$

$\hat{\omega} = \frac{2\pi f}{f_s} = \frac{2\pi \cdot 50}{75} = \frac{100}{75}\pi = \frac{4}{3}\pi$

$x[n] = 4\cos(2\pi(50)\frac{n}{75} + \frac{\pi}{3})$



(d)



(e) $f_r = 25\text{Hz}$